Unit 2:Data Condensation and Graphical Methods

Introduction:-
The information collected for any statistical investigation is called raw data. Raw data is voluminous i.e. it contains large information about qualitative and quantitative characteristics of the units in the population or sample. One can not able to assimilate all these things at a time; hence it is not possible to analyze the true behavior of the data. If one tries to analyze such house data, there are chances of drawing wrong conclusion about the nature of data. Thus there arises a need to reduce (condense) and simplify the raw data into such a form that its salient features may be brought out.

The technique used to condense and simplify the data is classification, tabulation and graphical representation of data. E.g. if we collect the data about percentage of marks of 1000 students, without lengthy scrutiny we can not answer the questions like how many students have obtained marks above 70%, between 60 to 70%, between 45 to 60% etc. To answer these questions, data may be classified according to % of marks.

*** Some terms**

Attributes: - Def. The characteristics which can not measured in units is called attributes (i.e. qualitative characteristics) E.g. intelligence, beauty, blood group, blindness, honesty etc.

Variable: - Def. The characteristics which can be measured in units and its value is changes is called variable (i.e. quantitative characteristics) E.g. Income, height, marks, weight, production of a factory etc.
There are two types of variables according to values taken by variable:-
i) Discrete variable:- A variable which takes only particular values or A variable which takes finite or countably infinite values is called discrete variable.
E.g. No. of childrens in a family, No. of workers in a factory, No. of accidents on highway in a year etc.
ii) Continuous variable: - A variable which takes all possible values between two appropriate values or A variable which takes uncountably infinite values is called continuous variable. E.g. age, income, weight, temperature etc.

Constant:- Def. The characteristics which can be measured in units and its value does not changes is called constant E.g. size of a room, total land of a country, gravitational force etc.

*** Classification:-** Def. The process of arranging the data into different classes (groups) according to similarities is called classification.
E.g. distribution of patients in to different wards according to their disease.

 # Objects of classification-
1) It condenses the data.
2) It simplifies the complex nature of data.
3) It helps in comparison with other data.
4) It shows prominent features of data.
5) It helps in further analysis.
6) It helps in drafting report.

***General principles of classification of raw data-**

1) It should not be ambiguous – classes should be rigidly defined. so that there should not any class for confusion regarding the placement of observation in the given classes.

2) It should be exhaustive and mutually exclusive – the classes should be exhaustive in the sense that each and every unit in the data must belongs to one of the classes. Also the classes are mutually exclusive so that any unit belongs to one and only one class i.e. classes are not overlapping.

3) It should be stable – for meaningful comparison of results the classification must be stable i.e. same pattern of classification should be adopted throughout the analysis and for further enquiries on the same subject.

4) It should be suitable for purpose – classification must be done by keeping objectives of the enquiries.

5) It should be flexible in the sense that it should be adjustable to new changed situations and conditions.

*** Classification by Attributes:-**

When a characteristics under consideration is an attribute, the simplest way of classification is to put all items possessing the attribute in one class and remaining items in other class. This type of classification in which whole data is classified in to only two classes according to possession of an attribute is called 'dichotomous 'or simple classification. e.g. students in a class are classified as male or female, pass or fail, rural or urban etc.

If we have more than two classes according to an attribute, such a classification is called 'manifold' e.g. The attribute blood group. we have more than two classes as A^{+ve}. B^{+ve} , O^{+ve} , AB^{+ve} , A^{-ve} B^{-ve} , O^{-ve} , AB^{-ve}

Whatever may be way of classification. the classes should be mutually exclusive and exhaustive .

*** Classification of variable** :-

When a characteristics under consideration is a variable, the classification is done according to values of variable. If variable is discrete, then each particular value of variable is considered as class e.g. No. of childrens in a family may be 0, 1. 2. 3. 4 etc. here each value forms a class and classification is done. If variable is continuous, then variable can be classified in to different intervals and these intervals are considered as class e.g. No. of childrens in a family can be classified according age as 0-5, 5-10, 10-15, 15-20 etc.

*** Some terms:-**

Class limits – Def. When data are classified in to different classes, the lowest and highest values of that class are called class limits of that class. The lowest value is called lower limit and highest value is called upper limit of that class.

Class width- **Def.** The difference between upper limit and lower limit of a class is called class width of that class.

Mid point of a class - **Def.** half of the sum of upper limit and lower limit of a class is called mid point of that class.

Frequency - **Def.** A number showing how many times a particular item repeated in the data is called frequency of that item.

Class frequency – **Def.** A number showing how many items belongs to a particular class is called class frequency of that class.

***Types of classes-**

There are two types of classes according to class limits.

i) Inclusive classes – These types of classes are so formed that both the limits (upper and lower) of a class are included in the same class. In this case variable under consideration is discrete.

E.g.

class	frequency
1 – 5	4
6 - 10	9
11 - 15	15
16 - 20	12
21 - 25	7

ii) Exclusive classes - In this type of classes the upper limit of a class is not included (excluded) in that class, but it is included in the next class. These types of classes are so formed that upper limit of a class is the lower limit of the next class. In this case variable under consideration is continuous.

E.g.

class	frequency
0 – 5	4
5 - 10	9
10 - 15	15
15 - 20	12
20 - 25	7

 # Conversion of inclusive classes into exclusive classes –

When inclusive type classes are given and we have to convert it into exclusive type classes, first find correction factor (c.f.) as follows.

c.f. = (lower limit of a class – upper limit of previous class) / 2

subtract this c.f. from lower limit and add to upper limit of each of the given inclusive type of classes, we get exclusive type classes

E.g. suppose inclusive classes are

class	frequency
1 – 5	4
6 - 10	9
11 - 15	15
16 - 20	12
21 - 25	7

Now, to convert it in to exclusive classes, we have to find correction factor

c.f. = (lower limit of a class – upper limit of previous class) / 2

$= (6 - 5) / 2 = 0.5$

Subtract 0.5 from lower limit and add 0.5 to upper limit of each of the given inclusive type of classes, we get exclusive type classes as follows.

class	frequency
0.5 – 5.5	4
5.5 – 10.5	9
10.5 – 15.5	15
15.5 – 20.5	12
20.5 – 25.5	7

Formation of classes when midpoints are given –

When midpoints are given and we have to form corresponding classes, first find correction factor (c.f.) as follows.

c.f. = (difference between two successive midpoints) / 2

subtract c.f. from midpoint to get lower limit and add c.f. to midpoint to get upper limit of the respective classes.

E.g. suppose midpoints of classes and respective frequencies are

Midpoint	frequency
10	4
15	9
20	15
25	12
30	7

Now, to form classes corresponding to given midpoints, we have to find correction factor.

c.f. = (difference between two successive midpoints) / 2

= (15 – 10) / 2 = 2.5

subtract 2.5 from midpoint to get lower limit and add 2.5 to midpoint to get upper limit of the respective classes as follows

Midpoint	Class	frequency
10	7.5 – 12.5	4
15	12.5 – 17.5	9
20	17.5 – 22.5	15
25	22.5 – 27.5	12
30	27.5 – 32.5	7

Open – end classes – Def. If lower limit of the lowest class or upper limit of the highest class or both are not specified, then the classes are called open – end classes.

E.g.

Class	frequency
Below -10	4
10 – 20	9
20 – 30	15
30 – 40	12
40 and above	7

Note – In case of open –end classes we can not find midpoints for first or last or both the classes, so to find midpoints make some assumption. If closed classes are of equal width assume same width for open –end classes and obtain midpoints.

*** Frequency distribution –** Def. A classification showing the different values of variable or classes for values of variable and their respective frequencies side by side is called frequency distribution.
OR The way by which the frequencies are distributed over different values or classes of values of variables is called frequency distribution.
There are two types of frequency distribution according to variable under consideration is discrete or continuous.

i) Discrete frequency distribution (simple or ungrouped) –
A frequency distribution in which different values of variable and their frequencies are shown side by side is known as discrete frequency distribution. In this case variable under consideration is discrete.
While formation of discrete frequency distribution we have to just count how many times a particular value (item) repeated in the data, this number is the frequency of that value. The work of counting is made easy by using tally bars.
E.g. suppose we have a data showing number of printing mistakes per page of a book containing 45 pages 2,4,1,5,0,0,2,0,4,3,2,1,1,0,4.0.2.1.3.1.5.0.3.2.0.4.2,1.5.1,0,2,1,3,4,1,0,3,4.5.2.1.0,1,1.
Note that here variable under consideration is No. of printing mistakes (X), it is discrete and different values taken by it are 0,1,2,3,4,5. The discrete frequency distribution formed is as follows.

X	Tally bars	F
0	~~||||~~ ~~||||~~	10
1	~~||||~~ ~~||||~~ ||	12
2	~~||||~~ |||	8
3	~~||||~~	5
4	~~||||~~ |	6
5	||||	4

ii) Continuous frequency distribution (grouped) –
A frequency distribution in which different classes for values of variable and their frequencies are shown side by side is known as continuous frequency distribution. In this case variable under consideration is continuous.
While formation of continuous frequency distribution we have to count how many values (items) belongs to particular class, this number is the frequency of that class. The work of counting is made easy by using tally bars.
E. g suppose we have data showing % of marks obtained by 65 students 60.15, 55.40, 45.45, 52.00, 52.60, 36.00, 45.00, 63.25, 44.50, 71.40, 60.50. 48.52. 65.60, 63.25. 74.00, 49.32, 62.45, 67.52, 69.50, 57.60, 62.55. 73.18, 56.00, 48.55, 59.32, 43.81, 56.12, 40.00, 64.25. 57.60, 47.52, 51.00, 56.00, 53.45, 65.00, 58.00, 63.45, 48.23, 57.20, 56.23, 50.0. 58.12, 53.36. 62.00, 57.12, 54.00, 59.50, 67.42, 57.00, 46.50, 58.00, 41.50, 62.18, 42.32, 64.00. 51.00, 54.18, 53.00. 68.12, 56.50, 37.25. 61.52, 57.00, 50.54, 47.12, 55.00, 58.00, 57.24, 63.00, 70.25.
Note that here variable under consideration is % of marks, it is continuous. The minimum and maximum value of variable is 36 and 74, so the classes formed are as 35-40, 40-45, ... 70-75.
The continuous frequency distribution formed is as follows.

class	Tally bars	F
35 - 40	||	2
40 – 45	~~||||~~	5
45 – 50	~~||||~~ ||||	9
50 – 55	~~||||~~ ~~||||~~ |	11
55 – 60	~~||||~~ ~~||||~~ ~~||||~~ ~~||||~~	20
60 – 65	~~||||~~ ~~||||~~ |||	13
65 – 70	~~||||~~ |	6
70 – 75	||||	4

General guidelines for forming class- intervals-

i) As far as possible the intervals should be of uniform width.
ii) The No. of classes should not be too small or too large.
iii) There should not be any class without any observation belonging to it.
iv) As far as possible avoid open-end classes.

* Cumulative frequency distribution-

A frequency distribution simply tells us that how many items belongs to a particular class. It does not answer the questions directly like, how many items have value less than or greater than a particular point. To answer this types of questions we have to obtain cumulative frequencies. Cumulative frequency is obtained by taking cumulative (progressive) sum of frequencies. There are two types of cumulative frequencies according to how the progressive sum is taken.

i) Less than type cumulative frequency (l.c.f.)-

L.c.f. of a class is a number which shows how many items have value less than or equal to the upper limit of corresponding class. L.c.f. is obtained by computing cumulative (progressive) sum of frequencies from lowest class to highest class.

ii) Greater than type cumulative frequency (g.c.f.)-

G.c.f. of a class is a number which shows how many items have value greater than or equal to the lower limit of corresponding class. G.c.f. is obtained by computing cumulative (progressive) sum of frequencies from highest class to lowest class.

E.g.

Class (marks)	F (No. of students)	lcf	gcf
10-20	5	5	81
20-30	9	14	76
30-40	13	27	67
40-50	19	46	54
50-60	15	61	35
50-70	11	72	20
70-80	6	78	9
80-90	3	81	3

Note- Less than cumulative frequency distribution can be written as.

Marks	No. of students (lcf)
Less than-20	5
Less than-30	14
Less than-40	27
Less than-50	46
Less than-60	61
Less than-70	72
Less than-80	78
Less than-90	81

Greater than cumulative frequency distribution can be written as.

Marks	No. of students(gcf)
Greater than-10	81
Greater than-20	76
Greater than-30	67
Greater than-40	54
Greater than-50	35
Greater than-60	20
Greater than-70	9
Greater than -80	3

*** Relative frequency** –

The ratio of class frequency to the total frequency is called relative frequency of that class.

i.e. relative frequency = (class frequency / total frequency) = (f / N)

Note that when we have to compare two frequency distributions and suppose that the total frequency of both the distributions is not same, the comparison becomes very difficult even though impossible. In n such situations relative frequency plays an important role in comparison.

E.g.

Class	F	Relative freq. (f/N)
10-20	5	5/81
20-30	9	9/81
30-40	13	13/81
40-50	19	19/81
50-60	15	15/81
50-70	11	11/81
70-80	6	6/81
80-90	3	3/81
Total	81	1

*** Graphical Representation of frequency distribution-**

We know that classification condense the raw data, it reveals the pattern of data and able us to make comparison. But when frequency distribution table is large in size comparison becomes difficult. Another way of achieving same objects perhaps better and more beautiful way is to represents the data by graphical method.

Graphs are easy to understand, present the facts in attractive manner and facilitate comparison hence conclusions are drawn quickly.

We have to study following graphs.

i) Histogram ii) Frequency polygon iii) Frequency curve iv) Ogives

1) Histogram:-

The simplest and popularly used way of representing grouped frequency distribution is to draw histogram. A histogram is defined as "the pictorial representation of grouped frequency distribution by means of adjacent rectangles whose areas are proportional to the frequencies of corresponding classes.

Construction of histogram – To construct histogram take class limits along X-axis and frequencies along Y-axis. Erect a rectangle on class-interval such that the area of rectangle is proportional to frequency of corresponding class. Since base of rectangle is the class width, so there is slight difference in the procedure of construction of histogram when the classes are of equal width and of unequal width.

i) When classes are of equal width – In this case width of all rectangles is same and is equal to width of class intervals. The height of rectangle is taken equal to frequency of respective class and hence the area of rectangle is proportional to respective class frequency.

ii) When classes are of unequal width – In this case width of all rectangles is varied and is equal to width of respective class intervals. The height of rectangle is so adjusted that area of rectangle is proportional to respective class frequency. So height of rectangle is taken equal to frequency density of respective class.

Frequency density of a class = frequency of a class / width of same class

Note i) Since histogram have adjacent rectangles, therefore in case of inclusive classes, first convert given inclusive classes into exclusive classes and then draw histogram.

ii) When midpoints are given, first prepare corresponding classes and then draw histogram.

iii) Histogram can not be drawn for open-end classes without any assumption about class width of open-end classes.

iv) mostly values of frequency densities are in fraction, to get integer values proceed as height of rectangle = frequency density X minimum class-width

Ex1) Represent the following data by histogram and hence find model value.

Class	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90	90-100
Freq.	5	11	14	20	25	22	13	9	3

Ex2) Represent the following data by histogram and hence find model value.

Income (1000Rs)	5-10	10-15	15-20	20-25	25-35	35-45	45-60	60-80
No.of employees	8	15	27	30	44	32	30	28

Ex3) Represent the following data by histogram and hence find model value.

Midpoint	25	30	35	40	45	50	55	60	65
Freq.	3	9	14	21	25	22	13	9	3

Ex4) Represent the following data by histogram and hence find model value.

Class	15-20	22-27	29-34	36-41	43-48	50-55	57-62	64-69	70-75
Freq.	11	18	24	32	45	40	33	21	13

2)Ogives (cumulative frequency curve):-

A curve which represents the cumulative frequency distributions is called as Ogives.
Since there are two types of cumulative frequencies, we have two types of ogives

i)Less than ogive – This curve represents less than cumulative frequency distribution. To construct it take class limits along X-axis and lcf along Y-axis. Plot the points by plotting lcf against upper limit of class. The successive points are joined by smooth free hand curve. This curve starts from zero on Y-axis and lower limit of the lowest class. The nature of this curve is increasing type.

ii) Greater than ogive – This curve represents greater than cumulative frequency distribution. To construct it take class limits along X-axis and gcf along Y-axis. Plot the points by plotting gcf against lower limit of class. The successive points are joined by smooth free hand curve. This curve ends at zero on Y-axis and upper limit of the highest class. The nature of this curve is decreasing type.

Ex1) Draw less than and greater than ogives for the following data, hence obtain median value.

Class	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90	90-100
Freq.	5	15	22	31	38	25	13	9	3

Ex2) Draw less than ogives for the following data, hence obtain quartiles.

Class	7-14	14-21	21-28	28-35	35-42	42-49	49-56	56-63	63-70
Freq.	8	12	22	30	38	27	18	9	6

*** Diagrammatic Representation of Data –**

When the data is classified according to an attribute, it can not be represented by graphical method. E.g. Population of country is classified according to states, this data can not be represented by graph. But such data can be represented by diagram very attractively.

The main difference between graphs and diagram is – Graph paper is necessary for construction of graph and there is mathematical relationship two variables under study. While diagrams are constructed on plane paper also. it is used for comparison only and not for mathematical relationship.

We have to study following diagrams.

i) Pie diagram

Pie diagram -

A pie diagram is a circle (represents the whole data) sub-divided into sectors by radii in such a way that the area of the sectors is proportional to the magnitude of the component items under study.

To construct pie diagram go through following steps –

i) Compute the angle representing each component by formula

Angle for component = (magnitude of component / total magnitude) x 360

ii) Draw a circle of any radius.

iii) Draw any radius (perfectly horizontal), consider this radius as a base line and draw an angle representing the first component. The sector so obtained will represent the proportion of first component. Now consider this newly drawn line (radius) as a base line and draw angle representing the second component. Proceed in this way until all component are shown.

Note- if some of the angles obtained are non-integer then round them in nearest integer.

Ex1) Following data shows expenditure of family over different heads per month. Represent the data by pie diagram and give title.

Item	Expenditure per month
Food	644
Clothing	200
House rent	420
Medicine	80
Others	96

Ex2) Following data shows cost of construction of house. Represent the data by pie diagram and give title.

Item	% of cost of construction
Labour	25
Bricks	15
Cement	20
Steel	15
Timber	10
Supervision	15

Assignment No.-2

Q1. Define – attribute, variable, discrete variable, continuous variable, class-limits, frequency, class-frequency, open-end classes, relative frequency, classification, l.c.f., g.c.f..

Q2. Define classification and state its objectives..

Q3. Explain frequency distribution.

Q4. Explain cumulative frequency distribution.

Q5. Explain construction of histogram.

Q6. Explain construction of ogives.

Q7. Explain construction of Pie-diagram.

Q8. Choose most correct alternative-

1. The characteristic which cannot measured in unit is called ....

a)attribute b)discrete variable c)continuous variable d)none of these

2. A number showing how many times a particular item repeated in the data is called............

a)frequency b)L.c.f. c)G.c.f. d)relative frequency

3. In..............types of classes both the limits of a class are included in the same class.

a) inclusive b) exclusive c) open-end d)all of these

4. In ....types of classes upper limit of a class is the lower limit of next class.

a) inclusive b) exclusive c) open-end d)all of these

5. if upper limit of the highest class is not specified, then the classes are of the type........

a) inclusive b) exclusive c) open-end d)all of these

6. A number showing how many items have value less than or equal to upper limit of the respective class is called ....

a)frequency b)L.c.f. c)G.c.f. d)relative frequency

7. G.c.f. can be obtained by computing cumulative sum of frequency from..................

a) lowest class to highest class b) highest class to lowest class c) (a) &(b) both d) none of these

8. While construction of histogram In case of unequal classes the height of rectangle is taken equal to....... of the respective class.

a)frequency b) l.c.f c)g.c.f. d) frequency density

9. While constructing less than ogive lcg is plotted against................ of respective class.

a)lower limit b)upper limit c)midpoints d) none of these

10. .... is used for comparison only and not for mathematical relationship.

a) histogram b) ogives c)pie-diagram d) all of these.

11. Histogram represents ---------------

a) continuous frequency distribution b) less than cumulative frequency distribution

c) relative frequency distribution d) none of these

12. Ogives represents --------------

a) continuous frequency distribution b) less than cumulative frequency distribution

c) relative frequency distribution d) none of these

13. The sum of relative frequencies is ------------ always.

a) any real number b) any positive integer number c) 1 d) none of these

14. The difference between upper and lower limits of a class is called ------ of that class.

a) class width b) midpoint c) relative frequency d) none of these

15. To draw frequency curve, frequency is plotted against ------------- of the corresponding class.

a) upper limit b) lower limit c) midpoint c) none of these

16. Frequency curve shows asymptotic nature along X-axis, if the classes are of --------- type.

a) inclusive b) exclusive c) open-end d) all of these

17. Classification ---------------

a) condense the raw data b) simplifies the complex nature of data

c) helps in drafting the report. d) all of these

18. A discrete variable takes ---------------- values.

a) finite b) particular c) countable infinite d) all of these

19. the progressive sum of frequencies from lowest class to highest class provides -------------

a) relative frequency b) l.c.f. c) g.c.f. d) frequency density